

CSE 719: Distributed & Cloud Computing

Lecture 1 — Distributed System Fundamentals

Model Answers: 2020 & 2021 Papers

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Tanenbaum/Sinha

Contents

1	2020 Examination (CSE-813, 52.5 marks)	2
1.1	Q1(a) — Define distributed and cloud systems; give examples [2+1 marks]	2
1.2	Q1(b) — Tightly vs loosely coupled systems [3.25 marks]	2
1.3	Q1(c) — Distributed vs parallel processing [2.5 marks]	3
1.4	Q2(a) — Location, relocation, and migration transparency [3 marks] .	3
1.5	Q2(c) — Design principles for better performance [3 marks]	4
2	2021 Examination (CSE 813, 52.5 marks)	4
2.1	Q1(a) — Define a distributed system; real-time examples [1.75 marks] .	4
2.2	Q1(b) — Advantages over standalone; disadvantages [2.5 marks] . .	5
2.3	Q1(c) — Scalability + challenges [2.5 marks]	5
2.4	Q1(d) — Marshalling / Unmarshalling; ACID properties [2 marks] . .	6
3	Quick Summary — Exam Patterns & Must-Know	6

2020 Examination (CSE-813, 52.5 marks)

Q1(a) — Define distributed and cloud systems; give examples [2+1 marks]

Define distributed and cloud systems. Give appropriate examples.

Distributed System (Tanenbaum): “A collection of independent computers that appears to its users as a single coherent system.”

- Each node is **autonomous, asynchronous, and failure-prone**; they communicate via an **unreliable network**.
- *Examples:* the World Wide Web, DNS, NFS (file system), BitTorrent (P2P), Google Search.

Cloud System: A model for **on-demand, network access to a shared pool of configurable computing resources** (servers, storage, applications) that are rapidly provisioned and released, billed per use.

- *Examples:* Amazon EC2/S3 (IaaS), Google App Engine (PaaS), Microsoft Office 365 (SaaS).

Relationship: A cloud is a large-scale distributed system delivered as a metered utility over the internet.

Q1(b) — Tightly vs loosely coupled systems [3.25 marks]

Differentiate tightly and loosely coupled systems with appropriate figures.

Criterion	Tightly Coupled	Loosely Coupled
Memory	Shared global memory	Private local memory per node
Communication	Via shared memory reads/writes	Via message passing over network
Clock	Common/synchronised clock	Independent clocks
Coupling	High — fast, low-latency	Low — scalable, geo-distributed
Failure model	One failure can affect all	Nodes fail independently
Example	SMP multiprocessor	Distributed system / cluster

Figures (draw in exam):

- **Tightly coupled:** CPU₁, CPU₂, CPU₃ all connected via a common bus to one shared memory block.
- **Loosely coupled:** Each node has CPU + its own memory; nodes linked by a communication network (LAN/WAN).

Q1(c) — Distributed vs parallel processing

[2.5 marks]

In what respect are distributed computing systems better than parallel processing systems?

- **Scalability** — can scale to geographically dispersed, virtually unlimited nodes; parallel systems are bounded by a single machine's memory bus.
- **Fault tolerance** — no single point of failure; a node crash need not halt the system. In shared-memory parallel systems a fault is often fatal.
- **Resource & information sharing** — remote files, printers, and databases shared across sites.
- **Cost-effectiveness** — built from cheap commodity machines instead of expensive tightly-coupled hardware.
- **Geographic distribution** — services placed near users, reducing latency; parallel processing is always co-located.

Q2(a) — Location, relocation, and migration transparency

[3 marks]

How are location, relocation, and migration transparencies different from each other in distributed computing? Explain with examples.

Type	What it hides	Example
Location	<i>Where</i> a resource physically resides	A URL/DNS name gives no hint of the server's geographic site.
Migration	That a resource has <i>moved</i> to a new location (between client accesses) — name is unchanged	A file is silently migrated to another server; clients use the same path.
Relocation	That a resource <i>moved while actively in use</i>	A mobile user's session continues uninterrupted while the backend server changes.

Key distinction: **Location** = you cannot see where it is. **Migration** = it moved

between your accesses (name unchanged). **Relocation** = it moved *during* your active connection.

Q2(c) — Design principles for better performance [3 marks]

What are the design principles considered useful for better performance in a distributed system?

1. **Minimise communication** — batch messages, reduce round trips; exploit data locality.
2. **Caching and replication** — keep frequently accessed data close to the consumer to reduce remote reads.
3. **Exploit parallelism / concurrency** — overlap I/O and computation; use pipelining.
4. **Move computation to data** — when data is large, send the program to the data node rather than the reverse.
5. **Avoid centralised bottlenecks** — no single server or coordinator that every request must pass through; use decentralised/peer algorithms.
6. **Load balancing** — distribute work evenly so no node is idle while others are saturated.

2021 Examination (CSE 813, 52.5 marks)

Q1(a) — Define a distributed system; real-time examples [1.75 marks]

Define a distributed system. Mention a few real-time examples of distributed systems.

Definition: A collection of independent computers that appears to its users as a **single coherent system**, where entities communicate by message passing over an unreliable network.

Real-time examples:

- Air-traffic control systems
- ATM / inter-bank banking networks
- Online multiplayer games

- Sensor networks (IoT)
- Telecom switching networks

Q1(b) — Advantages over standalone; disadvantages [2.5 marks]

Mention the advantages of a distributed computing environment over standalone applications. What are the disadvantages of distributed systems?

Advantages:

- **Resource sharing** — hardware, software, and data shared across sites.
- **High reliability / fault tolerance** — redundancy; no single point of failure.
- **Scalability** — add nodes incrementally without replacing the whole system.
- **Cost-effective** — commodity hardware + shared infrastructure.
- **Geographic distribution** — services closer to users $\Rightarrow$ lower latency.

Disadvantages:

- **Complexity** — no global clock; harder to design, test, and debug.
- **Network dependence** — congestion/failure degrades or stops the system.
- **Security** — larger attack surface; data transits public/untrusted networks.
- **Consistency problems** — concurrent access to shared data requires synchronisation.

Q1(c) — Scalability + challenges [2.5 marks]

Define scalability. What are the challenges involved in designing scalable distributed systems?

Scalability: The ability of a system to **handle increasing load** (more nodes, users, or geographic spread) **without significant loss of performance**.

Three dimensions: **size scalability** (more nodes/users), **geographic scalability** (wider area), **administrative scalability** (more organisations).

Challenges:

- **Centralised components** — a single server becomes a bottleneck under high load.
- **Centralised data** — a single database cannot serve billions of requests.
- **Centralised algorithms** — any algorithm requiring global state cannot scale.

- **Communication latency** — grows as nodes spread across WAN distances.
- **Replica consistency** — keeping distributed copies in sync is harder at scale.

Solutions: hiding latency (async/caching), partitioning (sharding), replication.

Q1(d) — Marshalling / Unmarshalling; ACID properties [2 marks]

Define Marshalling and Unmarshalling. What are the ACID properties?

Marshalling: The process of **packing data items into a flat byte stream** (external representation) suitable for transmission across a network. (Also called *serialisation*.)

Unmarshalling: The inverse — **disassembling the received byte stream** back into the original data structures at the destination.

ACID Properties (transactions):

Letter	Property	Meaning
A	Atomicity	All-or-nothing: a transaction either completes fully or leaves no trace.
C	Consistency	Takes the database from one valid state to another (no invariant is broken).
I	Isolation	Concurrent transactions do not interfere; result equals some serial execution.
D	Durability	Once committed, changes survive crashes (written to persistent storage).

Quick Summary — Exam Patterns & Must-Know

- **Q1 of Section A = Fundamentals (2020 & 2021).** Definition + examples open both papers — have them memorised cold.
- **2022 & 2024 papers have no Lecture-1 questions.** All coverage is from 2020 and 2021.
- **Transparencies (2020 Q2a):** distinguish *Location* vs *Migration* vs *Relocation* clearly — the examiner tests the differences, not just names.
- **Coupling (2020 Q1b):** question always says “with appropriate figures” — draw shared-memory vs private-memory diagrams explicitly.

- **Scalability (2021 Q1c):** three dimensions + the three centralisation pitfalls (component, data, algorithm) $\Rightarrow$ 5 marks in this style.
- **Marshalling + ACID (2021 Q1d):** two-parter for only 2 marks — keep each part to 2–3 lines.
- **Slide vs textbook:** DS definition/design-goals/hard-issues = Lecture-01 slides. Coupling, transparencies, ACID = Tanenbaum/Sinha textbook.